Ghislain THEUBO

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Professional Summary

Junior Embedded Software Engineer (Ensimag) with proven **C/FreeRTOS** development on MCU-class constraints at **INRIA**: real-time control loops, sensor/comm pipelines, HW/SW bring-up with electrical partners, and rigorous validation. Comfortable with SPI/I2C/UART notions, Git, CI habits, and Python test scripts. Seeking **Embedded Software Engineer – Launcher** at Harmattan AI (Paris) — junior band of the 2–10y range — to own firmware on launchers/controllers under bare-metal/RTOS with strong test maturity path.

Core Strengths

Embedded C/C++ · **FreeRTOS / bare-metal notions** · Drivers & buses (SPI/I2C/UART) · Real-time control loops · Bring-up with HW · Unit/HITL testing mindset · Git/CI · English · Autonomy

Experience

Shift Technology – Platform Software Engineering — Paris, France
Software Engineer (CDI – Platform / Backend) — *Sept 2026 – Nov 2026*

- Contributed to the **platform / foundations** squad building shared services used by product teams: design and implementation of **C# / .NET** components and **high-performance APIs** on **Microsoft Azure**.
- Worked with cloud data and messaging building blocks (**Azure Cosmos DB**, **Service Bus**), configuration/transversal services, and infrastructure-as-code awareness (**Terraform**) in a hybrid SaaS environment.
- Practiced production engineering habits: design discussions, code reviews, monitoring mindset, incremental delivery, and collaboration with platform and product engineers.

Gustave Roussy & CentraleSupélec MICS Lab — Paris / Saclay, France
Software Engineer (CDD – Backend / Data Platforms) — *Jan 2026 – Sept 2026*

- Production Python services with tests, CI (GitHub Actions), Docker — software maturity habits transferable to firmware HITL/CI.
- PostgreSQL tuning and structured logging; clear documentation for operators of the system.

Dassault Systèmes – 3DEXPERIENCE / 3DSwym R&D — Vélizy-Villacoublay, France
Software Engineer Intern (Full-Stack Module Rewrite) — *Mar 2025 – Sept 2025*

- Large module delivery with TDD (∼**85%** coverage) and measurable performance (∼**35%**) — proof of quality discipline under industrial constraints.

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern – Real-Time Control Modules (Assistive Wheelchair) — *May 2024 – Aug 2024*

- Owned firmware-side **C/FreeRTOS** modules: scheduling, interrupts/timing awareness, sensor/comm pipelines on constrained hardware.
- Partnered daily with hardware-oriented teammates for integration and debug; iterative bring-up and field validation with end user.
- Improved pipeline responsiveness by ∼**40%**; documented code for reproducibility — prototype toward reliable deployment.

LIG – Laboratoire d'Informatique de Grenoble — Grenoble, France
Cloud Infrastructure Intern (OpenStack / Grid'5000) — *Feb 2025 – Mar 2025*

- Automation and measurement scripts — tooling mindset useful for bring-up and test benches.

Selected Projects

MCU FreeRTOS Control Pack | C, FreeRTOS, SPI/I2C/UART stubs — GitHub

- RTOS tasks for sensor poll + actuator loop; driver stubs from datasheet excerpts; unit tests on framing/CRC.
- Bring-up checklist + mock HITL harness — path toward production maturity.

INRIA FreeRTOS Product Path | C, FreeRTOS, Python — INRIA

- Real device, real user, measurable latency work — strongest signal for Launcher junior firmware role.

Bootloader / OTA Study Notes | C, flash layout concepts — GitHub

- Documented secure-update concepts (A/B slots, CRC, rollback) as ramp for launcher OTA responsibilities.

Technical Skills

Core: Embedded C/C++, FreeRTOS, MCU peripherals, SPI/I2C/UART, Git
Practices: unit tests, CI notions, bring-up scripts (Python), code review culture
Bonus path: RF framing lab, OTA/bootloader study notes, ruggedized system interest
Languages: English professional; French fluent

Education & Certifications

Grenoble INP - Ensimag — Grenoble, France
Engineering Degree – Information Systems / Software Engineering (Double Degree) — *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé — Yaoundé, Cameroon
Master's Degree in Computer Engineering – Top 12% of class — *2019 – 2023*

Certifications: Programming Foundations – TDD (LinkedIn, 2025) · Primerli – Insurance Industry Foundations (2025)

Languages & Extra

English: bilingual / professional · **French:** fluent · **Spanish:** intermediate
Extra: tutoring · tennis, football